Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 3 Quiz 1 - Forces and Dynamics

Mock Exam F · Form 2: Advanced reasoning

Multiple Choice Questions: Select the best answer. (1 point each)

1. A 10 kg box slides right. A 90 N force acts rightward at 20° above horizontal. $\mu_k = 0.15$; contact is maintained and vertical acceleration is zero. Find horizontal acceleration. Use g = 9.8 m/s².

(A) 14.898 m/s²

(B) 3.724 m/s²

(C) 7.449 m/s²

(D) -7.449 m/s²

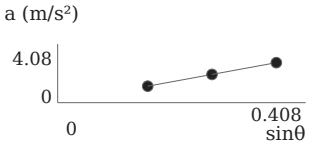
2. Release a block on a frictionless ramp and record downhill acceleration. Use g=10 m/s². Double block mass on the same frictionless ramp. Which X in the modified apparatus yields Y=2.88?

(A) 2.388

(B) -3.912

(C) 6.588

(D) 0.288



3. A 110 kg box initially at rest is pushed horizontally by 592.9 N. $\mu_s = 0.4$, $\mu_k = 0.25$. What is the acceleration just after slipping begins? Use g = 9.8 m/s².

(A) 2.94 m/s²

(B) -2.94 m/s²

(C) 5.88 m/s²

(D) 1.47 m/s²

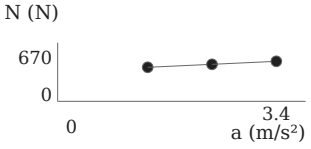
4. A 50 kg passenger stands on a scale in an elevator. Up is positive; use g=10 m/s². a (m/s²) -> N (N): (1.4, 570); (2.4, 620); (3.4, 670). Both sensors multiply their axes by the same unknown positive gain k. What is the uncorrected slope?

(A) 70

(B) 30

(C) 90

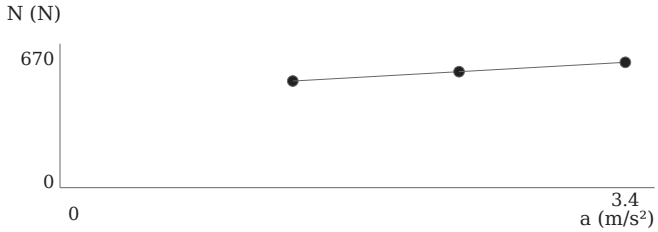
(D) 50



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. A 50 kg passenger stands on a scale in an elevator. Up is positive; use g=10 m/s². a (m/s²) -> N (N): (1.4, 570); (2.4, 620); (3.4, 670).

a. For the passenger on a scale setup, label the measured quantities and their reference directions or origins. **(1 pt)**



b. Double passenger mass in the same gravity. Predict the new slope and intercept and explain physically what changed. **(3 pt)**

6. The same two surfaces slide against each other. Vary the normal force and measure friction magnitude. N (N) -> fk (N): (1.4, 0.56); (2.4, 0.96); (3.4, 1.36).

a. Build the model for sliding-friction measurements from physics and predict both slope and intercept. **(2 pt)**

b. Switch to a surface with half the kinetic coefficient at the same normal loads. Predict the new slope and intercept and explain physically what changed. **(2 pt)**

7. Release a block on a frictionless ramp and record downhill acceleration. Use g=10 m/s². sinθ -> a (m/s²): (0.168, 1.68); (0.288, 2.88); (0.408, 4.08). Build the model for ramp tilt experiment from physics and predict both slope and intercept. **(3 pt)**